

# Assignment #3

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Example

Find the Fourier Integral representation of the function

$$f(x) = \begin{cases} 1 & \text{if } |x| < 1 \\ 0 & \text{if } |x| > 1 \end{cases}$$

$$A(\omega) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(v) \cos \omega v \, dv$$

$$= \frac{1}{\pi} \int_{-1}^1 \cos \omega v \, dv$$

$$= \frac{\sin \omega v}{\pi \omega} \Big|_{-1}^1 = \frac{2 \sin \omega}{\pi \omega}$$

$$B(\omega) = \frac{1}{\pi} \int_{-1}^1 \sin \omega v \, dv = 0$$

$$f(x) = \frac{2}{\pi} \int_0^{\infty} \frac{\cos \omega x \sin \omega}{\omega} \, d\omega$$

$$\int_0^{\infty} \frac{\cos \omega x \sin \omega}{\omega} \, d\omega = \begin{cases} \pi/2 & \text{if } 0 \leq x < 1 \\ \pi/4 & \text{if } x = 1 \\ 0 & \text{if } x > 1 \end{cases}$$



## 6.1) Laplace Transform

Example 1)

let  $f(t) = 1$  when  $t \geq 0$  Find  $f(s)$

Solution

$$\mathcal{L}(f) = \mathcal{L}(1) = \int_0^{\infty} e^{-st} dt$$

$$= \left. -\frac{1}{s} e^{-st} \right|_0^{\infty}$$

$$= \frac{1}{s}$$

$$\int_0^{\infty} e^{-st} f(t) dt = \lim_{T \rightarrow \infty} \int_0^T e^{-st} f(t) dt$$

$$\int_0^{\infty} e^{-st} dt = \lim_{T \rightarrow \infty} \left[ -\frac{1}{s} e^{-st} \right]_0^T$$

$$= \lim_{T \rightarrow \infty} \left[ -\frac{1}{s} e^{-sT} + \frac{1}{s} e^0 \right] = \frac{1}{s}$$



Example 2 Laplace Transform  $\mathcal{L}(e^{at})$  of the Exponential Function  $e^{at}$

let  $f(t) = e^{at}$  when  $t \geq 0$ , where  $a$  is a constant

Find  $\mathcal{L}(f)$

Solution

$$\mathcal{L}(e^{at}) = \int_0^{\infty} e^{-st} e^{at} dt$$

$$= \frac{1}{a-s} e^{-(s-a)t} \Big|_0^{\infty}$$

$$\mathcal{L}(e^{at}) = \frac{1}{s-a}$$

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2)  $t (0 < t < 2)$

In terms of unit step function

$$f(t) = t(u(t) - u(t-2))$$

Transform of the function

$$\begin{aligned} \mathcal{L}\{f(t)\} &= \mathcal{L}\{t(u(t)) - t(u(t-2))\} \\ &= \mathcal{L}\{t u(t)\} - \mathcal{L}\{t(u)(t-2)\} \\ &= \frac{1}{s^2} - e^{-2s} \left( \frac{1}{s^2} - \frac{2}{s} \right) \\ &= \frac{1}{s^2} - \frac{e^{-2s}}{s^2} + \frac{2e^{-2s}}{s} \end{aligned}$$



$$= \frac{1 - e^{-2s} + 2se^{-2s}}{s^2}$$

3)  $t-2 \ (t > 2)$

In terms of unit step function

$$f(t) = (t-2)u(t-2)$$

Transform of the function

$$\mathcal{L}\{f(t)\} = \mathcal{L}\{(t-2)u(t-2)\}$$

$$\mathcal{L}\{f(t-a)u(t-a)\} = e^{-as} \mathcal{L}\{f(t)\}$$

So,

$$= \frac{e^{-2s}}{s^2}$$

4)  $\cos 4t \ (0 < t < \pi)$

In terms of unit step function

$$f(t) = \cos 4t (u(t) - u(t-\pi))$$

Transform of the function

$$\mathcal{L}\{f(t)\} = \mathcal{L}\{\cos 2t u(t)\} - \mathcal{L}\{\cos 2t u(t-\pi)\}$$

$$= \frac{s}{s^2-4} - e^{-\pi s} \left( \frac{s}{s^2-4} \right)$$

$$= \frac{s}{s^2-4} (1 - e^{-\pi s})$$



Q)  $f(x) = x$   $(0 < x < 2\pi)$

let  $x = \frac{1}{2} a_0 + a_1 \cos x + a_2 \cos 2x + b_1 \sin x +$

$b_2 \sin 2x + \dots \rightarrow \textcircled{1}$

$$a_0 = \frac{1}{\pi} \int_0^{2\pi} f(x) dx$$

$$= \frac{1}{\pi} \int_0^{2\pi} x dx$$

$$= \frac{1}{\pi} \left[ \frac{x^2}{2} \right]_0^{2\pi}$$

$$= \frac{1}{\pi} \left[ \frac{(2\pi)^2}{2} - \frac{0}{2} \right]$$

$$= \frac{1}{\pi} \left[ \frac{4\pi^2}{2} \right]$$

$$\boxed{a_0 = 2\pi}$$



$$\sin n\pi = 0$$

$$a_n = \frac{1}{L} \int_{-L}^L f(x) \cos\left(\frac{n\pi x}{L}\right) dx$$

$$a_n = \frac{1}{1} \int_{-1}^0 (-1) \cos(n\pi x) dx + \int_0^1 (1) \cos(n\pi x) dx$$

$$a_n = -1 \int_{-1}^0 \cos n\pi x dx + \int_0^1 \cos n\pi x dx$$

$D' = n\pi$                        $D' = n\pi$

$$= \frac{-1}{n\pi} \int_{-1}^0 n\pi \cos n\pi x dx + \frac{1}{n\pi} \int_0^1 n\pi \cos n\pi x dx$$

$$= -1 \left[ \frac{\sin n\pi x}{n\pi} \right]_{-1}^0 + \left[ \frac{\sin n\pi x}{n\pi} \right]_0^1$$

$$= \frac{-1}{n\pi} \left[ (\sin n\pi(0)) - (\sin n\pi(-1)) \right] + \frac{1}{n\pi} \left[ (\sin n\pi(1)) - (\sin n\pi(0)) \right]$$

$$= \frac{-1}{n\pi} [0 - 0] + \frac{1}{n\pi} [0 - 0]$$

$$\boxed{a_n = 0}$$

$$b_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \sin\left(\frac{n\pi x}{\pi}\right) dx$$

$$= \frac{1}{\pi} \int_0^{2\pi} \overset{u}{x} \overset{v}{\sin nx} dx$$

$$v_1 = \int v dx$$

$$v_2 = \int v_1 dx$$

$$\int u \cdot v dx = uv_1 - u'v_2$$

$$b_n = \frac{1}{\pi} \left[ x \left( -\frac{\cos nx}{n} \right) - 1 \left( -\frac{\sin nx}{n^2} \right) \right]_0^{2\pi}$$

$$= \frac{1}{\pi} \left[ \frac{-x \cos nx}{n} + \frac{\sin nx}{n^2} \right]_0^{2\pi}$$

$$= \frac{1}{\pi} \left[ \frac{-2\pi \cos 2n\pi}{n} + \frac{\sin 2n\pi}{n^2} \right] - \left( \frac{0 \cos(0)}{n} + \frac{\sin 0}{n^2} \right)$$

$$= \frac{1}{\pi} \left[ \left( \frac{-2\pi(1)}{n} + \frac{0}{n^2} \right) - \left( \frac{0(1)}{n} + \frac{0}{n^2} \right) \right]$$

$$= \frac{1}{\pi} \left[ \frac{-2\pi}{n} \right]$$

$$\boxed{b_n = \frac{-2}{n}}$$



$$\therefore n = 1, 2, 3, \dots$$

$$b_1 = -2, \quad b_2 = -1, \quad b_3 = -\frac{2}{3}, \dots$$

Now Substituting the values of  $a$ 's and  $b$ 's

$$x = \frac{1}{2} (2\pi) - 2 \sin x - \sin 2x - \frac{2}{3} \sin 3x - \dots$$

$$x = \pi - 2 \left( \sin x - \frac{1}{2} \sin 2x + \frac{1}{3} \sin 3x - \dots \right)$$

Ans



$$Q2) f(x) = -1 \quad (-1 < x < 0), \quad f(x) = 1 \quad (0 < x < 1)$$

$$P = 2L = 2$$

$$2L = 2$$

$$L = \frac{2}{2} = 1$$

$$\text{let } f(x) = \frac{1}{2} a_0 + \sum_{n=1}^{\infty} \left\{ a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right) \right\}, \dots$$

$$a_0 = \frac{1}{L} \int_{-L}^L f(x) dx$$

$$a_0 = \frac{1}{1} \int_{-1}^0 (-1) dx + \int_0^1 (1) dx$$

$$a_0 = [-x]_{-1}^0 + [x]_0^1$$

$$a_0 = [(0) - (-1)] + [(1) - (0)]$$

$$a_0 = 1 + 1$$

$$\boxed{a_0 = 2}$$



$$\sin n\pi = 0$$

$$a_n = \frac{1}{L} \int_{-L}^L f(x) \cos\left(\frac{n\pi x}{L}\right) dx$$

$$a_n = \frac{1}{1} \int_{-1}^0 (-1) \cos(n\pi x) dx + \int_0^1 (1) \cos(n\pi x) dx$$

$$a_n = -1 \int_{-1}^0 \cos n\pi x dx + \int_0^1 \cos n\pi x dx$$

$D' = n\pi$                        $D' = n\pi$

$$= \frac{-1}{n\pi} \int_{-1}^0 n\pi \cos n\pi x dx + \frac{1}{n\pi} \int_0^1 n\pi \cos n\pi x dx$$

$$= -1 \left[ \frac{\sin n\pi x}{n\pi} \right]_{-1}^0 + \left[ \frac{\sin n\pi x}{n\pi} \right]_0^1$$

$$= \frac{-1}{n\pi} \left[ (\sin n\pi(0)) - (\sin n\pi(-1)) \right] + \frac{1}{n\pi} \left[ (\sin n\pi(1)) - (\sin n\pi(0)) \right]$$

$$= \frac{-1}{n\pi} [0 - 0] + \frac{1}{n\pi} [0 - 0]$$

$$\boxed{a_n = 0}$$



$$\cos 0 = 1$$

$$\cos n\pi = (-1)^n$$

$$\cos -n\pi = (-1)^{-n}$$

$$b_n = \frac{1}{L} \int_{-L}^L f(x) \sin \left( \frac{n\pi x}{L} \right) dx$$

$$= \frac{1}{1} \int_{-1}^0 (-1) \sin(n\pi x) dx + \int_0^1 (1) \sin(n\pi x) dx$$

$$= -\frac{1}{1} \int_{-1}^0 \sin(n\pi x) dx + \int_0^1 \sin(n\pi x) dx$$

$$= -\frac{1}{n\pi} \int_{-1}^0 n\pi \cdot \sin(n\pi x) dx + \frac{1}{n\pi} \int_0^1 n\pi \cdot \sin(n\pi x) dx$$

$$= -\frac{1}{n\pi} \left[ -\cos(n\pi x) \right]_{-1}^0 + \left[ -\frac{\cos n\pi x}{n\pi} \right]_0^1$$

$$= \frac{1}{n\pi} \left[ \cos(n\pi x) \right]_{-1}^0 + \frac{1}{n\pi} \left[ \cos(n\pi x) \right]_0^1$$

$$= \frac{1}{n\pi} \left[ (\cos(n\pi(0))) - (\cos(n\pi(-1))) \right] + \frac{1}{n\pi} \left[ (\cos n\pi) - (\cos n\pi(0)) \right]$$

$$= \frac{1}{n\pi} \left[ \cos(0) - (\cos(n\pi(-1))) \right] + \frac{1}{n\pi} \left[ (-1)^n - \cos 0 \right]$$

$$= \frac{1}{n\pi} \left[ 1 - (-1)^{-n} \right] + \frac{1}{n\pi} \left[ (-1)^n - 1 \right]$$



$$= \left[ \frac{1 - (-1)^{-n}}{\pi n} \right] - \left[ \frac{(-1)^n - 1}{\pi n} \right]$$

$$= \frac{(1 - (-1)^{-n}) - ((-1)^n - 1)}{\pi n}$$

$$= \frac{1 - (-1)^{-n} + 1 - (-1)^n}{\pi n}$$

$$b_n = \frac{2 - (-1)^{-n} - (-1)^n}{\pi n}$$

$$\therefore n = 1, 2, 3, \dots$$

$$n=1$$

$$b_1 = \frac{2 - (-1)^{-1} - (-1)^1}{\pi(1)}$$

$$b_1 = \frac{4}{\pi}$$

$$n=2$$

$$b_2 = \frac{2 - (-1)^{-2} - (-1)^2}{\pi(2)}$$

$$b_2 = 0$$

$$n=3$$

$$b_3 = \frac{2 - (-1)^{-3} - (-1)^3}{\pi(3)}$$

$$b_3 = \frac{4}{3\pi}$$

$$n = 4$$

$$b_4 = \frac{2 - (-1)^{-4} - (-1)^4}{\pi(4)}$$

$$b_4 = 0$$

$$n = 5$$

$$b_5 = \frac{2 - (-1)^{-5} - (-1)^5}{\pi(5)}$$

$$b_5 = \frac{4}{5\pi}$$

So,

$$f(x) = \frac{1}{2} a_0 + \sum_{n=1}^{\infty} \left\{ a_n \cos \left( \frac{n\pi x}{L} \right) + b_n \sin \left( \frac{n\pi x}{L} \right) \right\}, \dots$$

$$= \frac{1}{2} (2) + \sum_{n=1}^{\infty} \left\{ (0) \cos \left( \frac{n\pi x}{L} \right) + b_n \sin \left( \frac{n\pi x}{L} \right) \right\}, \dots$$

$$= \frac{1}{2} + \frac{4}{\pi} \sin \left( \frac{(1)\pi x}{1} \right) + \frac{4}{3\pi} \sin \left( \frac{3\pi x}{1} \right) + \frac{4}{5\pi} \sin \left( \frac{5\pi x}{1} \right) + \dots$$

$$f(x) = \frac{1}{2} + \frac{4}{\pi} \left( \sin \pi x + \frac{1}{3} \sin 3\pi x + \frac{1}{5} \sin 5\pi x + \dots \right)$$

Ans